



Multiple Choice:

1. D. P 124.59

Exact simple interest: Total number of days money is invested: 228

$$I = Pin$$

$$I = (2000)(0.10) \frac{228}{365}$$

$$I = 124.9$$

2. A. 7.86 years

$$F = P(1+i)^{2n}$$

$$2000 + 3000 = 2000 \left(1 + \frac{0.12}{2}\right)^{2n}$$

$$2.5 = 1.06^{2n}$$

$$\log 2.5 = 2n \log 1.06$$

$$n = 7.86 \text{ years}$$

3. B. P 371,287.13

$$ER_{\text{semi-annually}} = ER_{\text{quarterly}}$$

$$(1+i)^2 - 1 = \left(1 + \frac{0.08}{4}\right)^4 - 1$$

$$i = 0.0404$$

$$P = \frac{A}{i}$$

$$P = \frac{15,000}{0.0404}$$

$$P = 371,287.13$$

4. D. 11.60 % compounded monthly

Solving for the effective rates:

A. $ER = 12.35\%$ (if compounded annually)

B.

$$ER = (1+i)^m - 1$$

$$ER = \left(1 + \frac{0.119}{2}\right)^2 - 1 = 12.25\%$$

C.

$$ER = \left(1 + \frac{0.122}{4}\right)^4 - 1$$

$$ER = 12.77\%$$

D.

$$ER = \left(1 + \frac{0.116}{12}\right)^{12} - 1$$

$$ER = 12.24\%$$

5. C. P 791.26

$$d = \frac{(C_0 - C_n) i}{(1+i)^n - 1}$$

$$d = \frac{(10,000 - 500)(0.04)}{(1+0.04)^{10} - 1}$$

$$d = 791.26$$

6. A. 2,334

To avoid loss means that the minimum production must be enough to breakeven.

Let: x = number of cases to be sold each year to breakeven

$$\text{Income} = \text{Expenses}$$

$$65x = 50x + 35,000$$

$$x = 2,333.33 \text{ cases}$$

7. D. P 1,295.05

$$P = \frac{A[(1+i)^n - 1]}{(1+i)^n i}$$

$$10000 = \frac{A[(1+0.05)^{10} - 1]}{(1+0.05)^{10} (0.05)}$$

$$A = 1295.05$$

8. B. 8 %

$$d = \frac{FC - SV}{n}$$

$$d = \frac{100,000 - 20,000}{10}$$

$$d = 8,000$$

$$\text{rate} = \frac{d}{FC} \times 100\%$$

$$\text{rate} = \frac{8,000}{100,000} \times 100\%$$

$$\text{rate} = 8\%$$

9. C. 5 %

$$F = P(1+in)$$

$$5000 + 250 = 5000(1+i)$$

$$i = 5\%$$

10. B. Capital recovery

11. C. acid-test ratio

12. A. 20.15

$$F = P(1+i)^n$$

$$4P = 3,500 \left(1 + \frac{0.07}{2}\right)^{2n}$$

$$4 = (1.035)^{2n}$$

Take log on both sides:

$$\log 4 = \log (1.035)^{2n}$$

$$\log 4 = 2n \log 1.035$$

$$n = 20.15 \text{ years}$$

13. A. P 153

$$P = P_1 + P_2$$

$$P = \frac{F_1}{(1+i)^3} + \frac{F_2}{(1+i)^4}$$

$$P = \frac{100}{(1+0.08)^3} + \frac{100}{(1+0.08)^4}$$

$$P = 153$$

14. A. P 10,834.38

$$P = \frac{A[(1+i)^n - 1]}{(1+i)^n i}$$

$$P = \frac{2,000 \left[\left(1 + \frac{0.12}{4}\right)^6 - 1 \right]}{\left(1 + \frac{0.12}{4}\right)^6 i}$$

$$P = 10,834.38$$

15. C. P 13,000

$$i = \frac{0.12}{12}$$

$$i = 0.01$$

$$n = 12(2) = 24$$



$$F = \frac{A[(1+i)^n - 1]}{i}$$

$$F = \frac{500 \left[\left(1 + \frac{0.12}{12} \right)^{2(12)} - 1 \right]}{\frac{0.12}{12}}$$

$$F = 13,486.70$$

16. B. 25

Using Matheson formula:

$$k = 1 - \sqrt[n]{\frac{C_m}{C_0}}$$

Book value at the end of 10 years,
 C_{10} :

$$C_{10} = 40,545.73$$

Substituting in the formula:

$$k = 1 - \sqrt[10]{\frac{40,545.73}{720,000}}$$

$$k = 0.25$$

$$k = 25\%$$

17. C. P 146,000

$$C_0 = 500,000 + 30,000$$

$$C_0 = 530,000$$

$$C_n = 0.10(500,000)$$

$$C_n = 50,000$$

$$d = \frac{C_0 - C_n}{n}$$

$$d = \frac{530,000 - 50,000}{5}$$

$$d = 96,000$$

Let: C_4 = book value at the end of 4 years

$$C_4 = C_0 - d(m)$$

$$C_4 = 530,000 - 96,000(4)$$

$$C_4 = 146,000$$

18. A. sinking fund factor

19. D. P 666.39

Solving for the total number of days money was invested, d

January 15 – 31	= 16
February	= 29 (leap year)
March	= 31
April	= 30
May	= 31
June	= 30
July	= 31
August	= 31
September	= 30
October 1 – 12	= 12
	271 days

Substitute values in Eq. 1

$$I = 5,000(0.18) \left(\frac{271}{366} \right)$$

$$I = 666.39$$

20. P 467,760

$$P = \frac{A[(1+i)^n - 1]}{i(1+i)^n}$$

$$P = \frac{80000[(1+0.15)^{15} - 1]}{(0.15)(1+0.15)^{15}}$$

$$P = 80000(5.847)$$

$$P = 467,760 \rightarrow \text{Ans}$$

21. A. P 7,106.32

$$CR = \frac{(P - SV)[i(1+i)^n]}{(1+i)^n - 1} + i(SV)$$

$$CR = \frac{(50000 - 5000) \left[\frac{(0.08)}{(1+0.08)^{10}} - 1 \right]}{+0.08[0.1(50000)]}$$

$$CR = 7,106.32 \rightarrow \text{Ans}$$

22. B. P 322,549.33

Capitalized cost = CC = C_0 + present worth of all depreciation charges

Let P = present worth of all depreciation charges using sinking fund method

$$P = \frac{C_0 - C_n}{(1+i)^n}$$

$$P = \frac{300,000 - 30,000}{(1+0.18)^{15}}$$

$$P = 22,549.33$$

$$CC = 300,000 + 22,549.33$$

$$CC = 322,549.33$$

23. B. P 280,000

$$CC = FC + \frac{A}{i}$$

$$CC = 80,000 + \frac{12,000}{0.06}$$

$$CC = 280,000$$

24. C. P 9,250,000

Using Depletion Allowance Method:

$$\text{Dep. charge} = 0.22(50000000)$$

$$= 11000000$$

$$\text{Dep. charge} = 0.50(18500000)$$

$$= 9250000$$

Use the smaller value, P 9,250,000

25. D. P 262

Getting first the present worth of all money,

$$P = \frac{1000}{(1+0.10)^5} + \frac{2000}{(1+0.10)^{10}} + \frac{3500}{(1+0.10)^{15}}$$

$$P = 2229.88$$

Getting the equivalent annual cost,

$$A = \frac{2229.88(1+0.10)^{20}(0.10)}{((1+0.10)^{20} - 1)}$$

$$A = 262$$

END